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RESEARCH ARTICLE

Depth electrode recordings show double dissociation between pitch processing in lateral Heschl's gyrus and sound onset processing in medial Heschl's gyrus

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Abstract Pitch is a fundamental perceptual attribute of sounds. Our ability to discriminate, separate, and identify sounds relies heavily on pitch. Recent neuroimaging studies in humans have provided converging evidence for the existence of a "pitch center"—a region on the superior temporal plane (STP) lateral to Heschl's gyrus specialized in pitch extraction—but a direct confirmation is still missing. Intracerebral recordings in humans are ideally suited for such a confirmation. Here we report results from depth electrode recordings in a patient undergoing investigation for epilepsy. We demonstrate a double dissociation between responses from the medial and lateral STP around Heschl's gyrus to the onset of sound energy and the onset of pitch. Three pieces of evidence support this finding: (1) the response to sounds that do not contain pitch is small in the lateral STP compared to the medial STP; (2) sounds that contain pitch evoke a strong response in the lateral STP; (3) at the transition from noise to a specialised noise-like, but tonal, sound referred to as iterated ripple noise, where the onset of pitch is the sole acoustic event, only the lateral contact showed a response. Our results provide direct evidence for a pitch-specific area on lateral STP with intracranial recordings from the human auditory cortex.

Keywords Pitch · Auditory cortex · Human · Auditory evoked potentials

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Introduction

The perception of pitch is a fundamental function of the auditory system. Pitch is the perceptual attribute that underlies the perception of melody in music and prosody and speaker size in speech. Moreover, our ability to separate and identify concurrent sounds from different sources relies heavily on pitch. For pure-tones, pitch is directly related to frequency. For more complex sounds pitch has to be computed in the auditory brain. One example is the perceptual phenomenon of the missing fundamental, in which the pitch of a complex harmonic tone is heard at the fundamental frequency (lowest frequency of the harmonic series) even if that frequency itself is not present in the sound. Pattern-recognition theories of pitch perception were proposed to account for the missing fundamental phenomenon (Goldstein 1973). These theories assume that the pitch of a complex tone is extracted by a centrally located processor that measures frequency relationships between resolved spectral components. Pattern-recognition theories explain the perception of the missing fundamental, pitch percepts induced by inharmonic complex tones, and the pitch ambiguity of complex tones consisting of few harmonics. However, these theories cannot account for pitch percepts elicited by the periodicity of sounds in the absence of spectral cues. Licklider (1951) produced the first computational model of pitch perception based on time intervals within auditory frequency channels. The signal is filtered to simulate cochlear frequency channels; then, a running autocorrelation function is calculated in each channel to reveal any periodicity. This model has since been extended to enable quantitative predictions of pitch perception and it accounts successfully for a large number of pitch phenomena (Denham 2005; Meddis and O'Mard 1997, 2006; Wiegrebe et al. 2000; Yost et al. 1996). While aspects of these models

have been tied to the physiology of the auditory periphery, they do directly specify mechanisms of pitch representation in the cerebral cortex.

Several recent studies have addressed the question whether there is an area in the cerebral cortex that is specialized in the representation of pitch. Lesion studies suggested a special role of lateral Heschl's gyrus and immediately surrounding parts of the superior temporal plane (STP) it in the extraction of complex pitch (using sounds with a missing harmonic fundamental, Zatorre 1988) and pitch direction discrimination (Johnsrude et al. 2000). A series of neuroimaging studies has also provided evidence for the lateral Heschl's gyrus to be specifically involved the extraction of pitch (Griffiths et al. 1998; Patterson et al. 2002; Penagos et al. 2004). Activity in this area was shown to be modulated by the salience and value of pitch. Other studies have implicated this region in the integration of spectral information over time or frequency channels (Hall et al. 2002; Schönwiesner et al. 2005; Zatorre and Belin 2001), a function that seems crucial for computing the pitch of complex time-varying sounds. A complementary piece of evidence comes from a neurophysiological study by Bendor and Wang (2005) in marmoset monkeys. They demonstrated that neurons in a small region anterolateral to the medial primary field A1 represent the missing fundamental of harmonic stimuli. The location of this area relative to other auditory fields is similar to the location of the pitch sensitive region in the human brain.

The temporal characteristics of pitch-related responses in the human auditory cortex have been studied with magnetoencephalography (MEG). These recordings have isolated a transient component ("pitch onset response", Krumbholz et al. 2003) and a sustained component ("sustained pitch response", Gutschalk et al. 2002, 2004) of the auditory evoked field that indicate pitch processing. These responses have larger latencies than typical responses to the onset of a sound, indicating a hierarchical process in which pitch extraction follows sound onset-related processing, or a longer computation time for pitch extraction. Using dipole modelling, the sources of these responses were identified on lateral Heschl's gyrus, matching the results of the functional imaging studies. This convergence of neuroimaging and MEG data provides strong evidence for pitch processing in lateral Heschl's gyrus, but a direct confirmation is still missing. Intracerebral recordings in humans are ideally suited for such a confirmation, because this technique has high temporal and spatial resolution, making it possible to locate the cerebral sources of pitch-related responses and relate their temporal characteristics to previous MEG results.

Krumbholz and colleagues (2003) developed a continuous stimulation paradigm, which separates pitch-specific responses from the responses to the onset of sound energy. They used a specialized stimulus, referred to as iterated ripple noise (IRN, Patterson et al. 1996, 2002; Yost, 1996a, b), the pitch and pitch strength of which can be varied without introducing any major spectral changes. The IRN was preceded by a portion of noise without pitch so that the transient response evoked by the sound onset had subsided before the pitch onset. It was thus possible to temporally isolate pitch-related responses from other processes.

Here we use the continuous stimulation paradigm as well as click and pure-tone stimulation with intracranial recordings from a patient's auditory cortex to provide direct evidence for a specialised role of lateral Heschl's gyrus in pitch processing. In particular, we show a double dissociation of sound onset responses in medial Heschl's gyrus and planum temporale and pitch-specific responses in lateral Heschl's gyrus and superior temporal gyrus.

Materials and methods

Patient

A 33-year-old male patient with medically refractory focal seizures undergoing investigation for epilepsy with intracranial and depths electrodes gave written informed consent to participate in this study. The patient's left cerebral hemisphere showed global atrophy without focal lesions or cortical dysplasia. He showed cranio-facial hypertrophy on the left and hemiconvulsion of the right hemibody. The patient had had epileptic seizures from infancy, but his family has no history of seizures or other neurological disorders. He is left-handed, and a Wada test revealed right hemisphere lateralization for speech processing. The patient was first implanted with frontal and temporal depth electrodes and centroparietal epidural electrodes at age of 21. The examination indicated left temporal epilepsy. He then underwent two surgical resections of epileptogenic brain tissue, a transcortical selective amygdohippocampectomy on the left at the age of 21, and, at the age of 24, an extension of the resection to include the left anterior temporal lobe below the superior temporal gyrus back to the level of the precentral sulcus (about 45 mm at the base of the superior temporal gyrus), as well as hippocampal and parahippocampal tissue. Seizures returned about one year after both surgeries and a third examination with epidural and depth electrodes was performed. At this time the patient agreed to participate in the present study, and sound-evoked potentials were recorded from the implanted electrodes. The Montreal Neurological Institute and Hospital Research Ethics Committee approved the procedures used in this study.

Stimuli

The patient was presented with a variety of sounds. Only the stimulation with clicks, pure-tones, noise and IRN is described here. The intensity of the stimulation was adjusted to a comfortable hearing level of 65 dB SPL. The patient first listened passively to a series of 150 monaural clicks, presented with 800 ms stimulus onset asynchrony. Then, sinusoids of 8 frequencies were played: 118, 200, 339, 574, 972, 1,646, 2,788, 4,723, and 8,000 Hz. The tones were 100 ms long and were gated on and off with 5 ms cosine-squared ramps. The tones were presented 100 times in pseudorandom order with a stimulus onset asynchrony of 800 ms. Finally, noise stimuli were played, consisting of 1 s of white noise followed by 500 ms of IRN. They were presented 100 times in pseudorandom order with a stimulus onset asynchrony of 2 s. The IRN was generated by delaying a copy of a noise by 8 ms and adding it back onto itself ("add-original" procedure, Yost et al. 1996). This process was repeated 16 times, leading to a robust apparent pitch of 125 Hz. The stimuli were then high-pass filtered at 500 Hz. Low-pitched IRN filtered in this way effectively excites all frequency channels in the same way as random noise (Patterson et al. 1996), thus making the onset of pitch the sole acoustic event at the transition from noise to IRN. Both portions were gated on and off with 5 ms cosine-squared ramps. The ramps overlapped at the transition, so that the envelope of the composite stimulus remained flat.

Data acquisition

The implantation consisted of a combination of depth electrodes and epidural electrodes according to the method of Olivier and colleagues (1994). Electrodes were implanted stereotactically using an image-guidance system (SSN Neuronavigation System, Mississauga, ON, Canada) through percutaneous twist drill holes in the skull.

Intracranial depth electrodes were manufactured on site by wrapping 76 μm stainless steel wire around a 254 μm stainless steel central core. The wires were coated with Teflon except for regions where the insulation was stripped to form electrode contacts. In total, there were nine contacts on each depth electrode that were spaced along the length of the core wire at 5 mm intervals. The deepest contact (contact 1) was made from the tip of the core wire and had an uninsulated length of 1 mm, while more superficial contacts (contacts 2–9) were formed from stripped sections of the marginal wire that was tightly wound to make 0.5 mm long coils. The effective surface area for each of the eight superficial contacts was 0.80 mm^2, and was 0.85 mm^2 for the single deep contact. The depth electrode in the superior temporal lobe, which was recorded from in the present study, was directed within the lower bank of the Sylvian fissure about 5 mm posterior from Heschl's gyrus, running parallel to it for its whole length. Five of the nine electrode contacts came to lie within brain tissue parallel and a few millimeters posterior to Heschl's gyrus (Fig. 1). The contacts have low electrical impedance and are sensitive to the local field potential within an area of about 10 mm radius. This estimate is based on experience with electrical stimulations of single electrode contacts during the pre-surgery procedure at the Montreal Neurological Institute. We verified this estimate empirically by computing cross-correlations of potential fluctuations during rest before the start of the experiment between all electrode contacts in all four depth electrodes. Cross-correlation fell below significance threshold for any two contacts further than 10 mm apart.

Three additional depth electrodes were placed, two orthogonally through the middle and posterior medial temporal gyrus, and one in the inferior parietal lobe. In addition, four epidural electrodes were situated overlying peri-sylvian and centro-parietal areas.

Recordings were referenced to an extra-dural lead. The EEG signal was band pass filtered between 0.1 and 70 Hz and digitally recorded with a sampling frequency of 200 Hz using a 128-channel Harmony system for long-term monitoring (Stellate, Montreal, QC, Canada).

The stimuli were played back at 48 kHz sampling frequency from a personal computer running Presentation (Neurobehavioural Systems, Albany, CA, USA), and presented via Etymotic ER2 insert earphones (Etymotic Research, Elk Grove Village, IL, USA; the patient was unable to wear circum- or super-aural headphones due to head bandages). One EEG channel was used to record trigger pulses sent by the stimulation software at every stimulus onset.

Data analysis

Data were read into Matlab (The MathWorks, Natick, MA, USA) using the Stellate Signal File Interface toolbox (Stellate, Montreal, QC, Canada) and analyzed with the EEGLAB toolbox (Delorme and Makeig 2004). Data points exceeding a max–min difference of 150 μV were considered artefactual and a 500-ms window surrounding the artefact was removed from subsequent analysis. The data were bandpass-filtered between 0.1 and 35 Hz. The data of experiment 1 (pure-tone stimulation) were divided into 700-ms stimulus-locked epochs covering the period from −100 to 600 ms relative to stimulus onset. The data of experiment 2 (IRN stimulation) were divided into 2,100-ms stimulus-locked epochs covering the period from −100 to 2,000 ms relative to stimulus onset. The epochs for each condition were averaged and baseline-corrected to the 100-ms period of silence just before stimulus onset.

Statistics of the responses were computed using standard resampling procedures. The reliability of the average

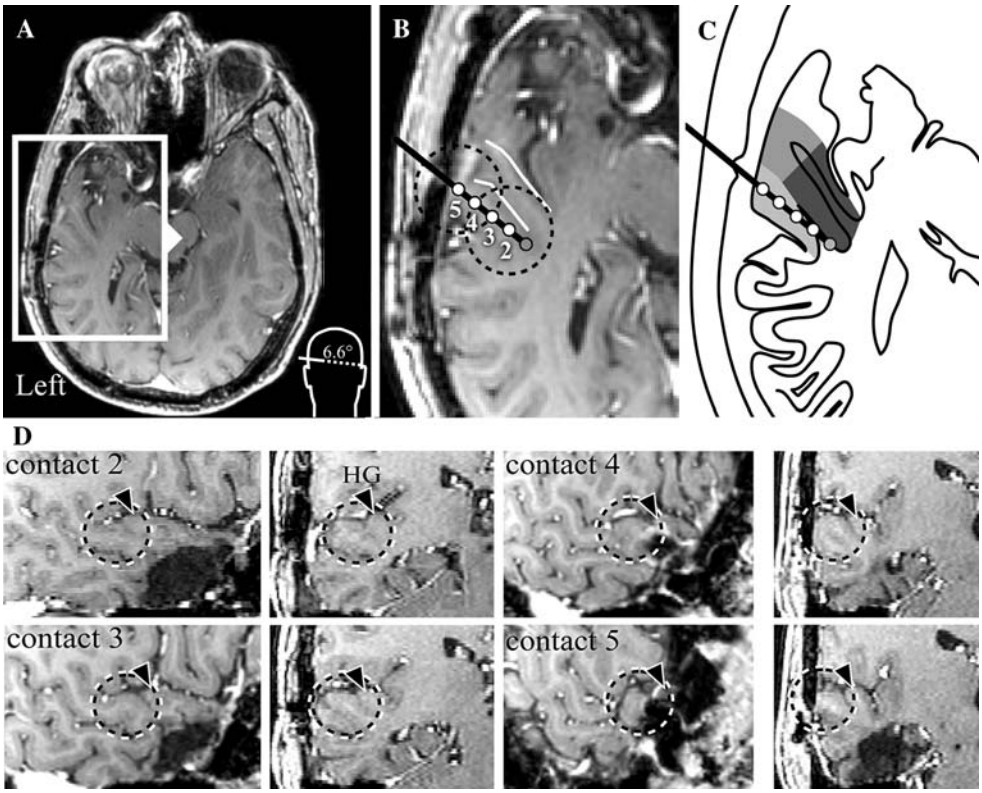

Fig. 1 **a** Structural image of the patient's brain, rotated into the axis of the implanted electrode (see *inset*), with the section enlarged in **b** highlighted. **b** Trajectory of the electrode (*black line*) in the superior temporal lobe. Of the five electrode contacts within grey or white matter, four (*white circles*) showed significant sound evoked responses (contacts *2–5*). Each contact is sensitive to the local field potential within a 10 mm radius, shown as *black dotted circle* for the medial (*2*) and the lateral contact (*5*). **c** Schematic outline of skull, grey and white matter with the electrode trace indicated (*thick black line*). The medial 2/3 of HG, a conventional estimate of primary auditory cortex, are marked in *dark grey*, the remaining lateral part of HG are marked in middle *grey*, and the planum temporale is marked in *light grey*. **d** Parasagittal and coronal slices through the location of the electrode contacts. The *dotted circles* indicate the active radius around each contact, and the *arrow* points to HG

event-related responses was estimated by computing 10,000 mean evoked responses generated by randomly drawing with replacement from the measured single trials. The bootstrap standard error of the mean was then calculated as the standard deviation of the distribution of resampled means. Significant differences between responses from different electrode contacts were found by randomly exchanging single trials between electrode contacts 10,000 times, each time calculating the new average response at each contact and the difference between the responses. If there is a significant difference, then the amplitude difference between the original responses will fall outside 95% of the distribution of resampled amplitude differences.

Results

Sound evoked responses were recorded from all contacts of the electrode in the superior temporal lobe, except the most medial (contact 1), where response amplitudes were not significant compared to baseline fluctuations (bootstrap test). The waveform and magnitude of the responses differed between different recording sites and stimuli.

Click stimulation

Clicks evoked strong bipolar responses in the medial contacts 2 and 3, and weaker responses in the lateral contacts 4 and 5 (Fig. 2b). The first stable component of the responses was a negative-going potential with amplitudes and latencies at the different sites from medial to lateral on the STP as follows: contact 2 −52.2 μV at 90 ms, contact 3 −33.7 μV at 115 ms, contact 4 −13 μV at 105 ms, and contact 5 −18.9 μV at 105 ms. This wave was followed by a positive-going potential with a peak latency of 235 ms, clearest in the two lateral contacts, and a sustained positive potential extending about 600 ms past stimulus offset. Responses from contact 4 were small in amplitude compared to the neighbouring contacts.

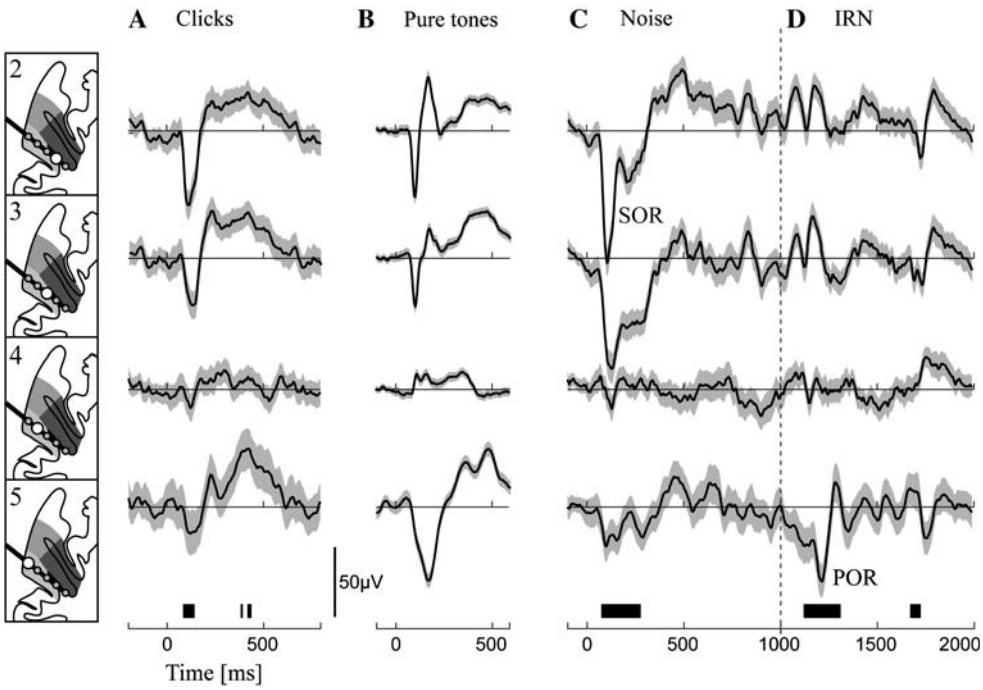

Fig. 2 *a–d* Responses obtained from the four electrode contacts to clicks, pure-tones (average of responses to all frequencies shown), and noise stimuli. The schematic outlines in the *left column* show the location of the contacts (*white circles*) with respect to anatomical markers (*grey* highlights as in Fig. 1c). The *dotted line* in *d* marks the onset of IRN pitch in the continuous noise stimulus. The *grey area* around the traces shows the bootstrap 95% confidence interval, and the *thick black lines* in the lowest panel highlight significant amplitude differences ($P < 0.05$, bootstrap test) between responses from contacts 5 and 2. *SOR* sound onset response, *POR* pitch onset response

Pure-tone stimulation

Responses at the medial contacts were strongly frequency dependent, whereas the lateral contact 5 showed very similar responses to all frequencies (Fig. 3). Responses from contact 4 were relatively small and inconsistent across frequencies. Figure 4 shows the peak-to-peak (max–min) amplitude within a time window from 50 to 200 ms past stimulus onset as well as the latency of the first negative deflection for all 8 frequencies. In the two medial contacts, response latencies decreased systematically from about 125 ms at 118 Hz to about 98 ms between 1 and 2.8 kHz. At 4.7 kHz response latency increased again to about 112 ms. At the lateral contact 5 there was a trend for the response latency to increase with frequency from about 160 ms at 118 Hz to 180 ms at 4.7 kHz.

Contact 2 responded strongest to the 972 Hz tone with a systematic decrease of the response amplitude toward lower and higher frequencies. The response preference appears to shift slightly toward lower frequencies at contact 3, where the 340 Hz tone elicited the strongest response. The shape of the responses changed dramatically from the medial to lateral contacts. Contacts 2 and 3 showed a clear biphasic response to most frequencies, with a negative-going potential peaking about 100 ms after stimulus onset, followed by a positive potential with about 180 ms peak latency. Contact 5 showed a broad negative-going potential with a latency of 180 ms. From about 350–600 ms after stimulus onset, a long sustained positive potential was recorded form medial and lateral contacts.

Noise and IRN stimulation

Recordings from the medial contacts 2 and 3 showed a clear response to the noise onset (Fig. 2d, contact 2 −101.8 μV at 105 ms, contact 3 −84.3 μV at 130 ms). There was no significant response to the onset of the pitch. The lateral contact 5 showed a small response (−30.4 μV at a 100 ms) to the noise onset, but a larger response to the onset of pitch of −57.4 μV amplitude at a latency of 220 ms after pitch onset. The amplitude of this response first reaches statistical significance against the baseline variance at 115.5 ms after pitch onset ($P = 0.05$, bootstrap test). There was thus a clear double dissociation between medial and lateral contact sites in their responses to sound onset and pitch onset. The significant differences between the responses from contacts 5 and 2 to clicks and noises are highlighted by thick black lines below the electrode traces in Fig. 2 ($P = 0.05$, bootstrap test).

Discussion

We recorded sound evoked potentials from a depth electrode in an epilepsy patient's auditory cortex. The results should be interpreted with caution due to the limitations inherent in experimenting with human patients. First, time constraints allowed us to collect only a limited number of responses and we therefore focus on robust long-latency potentials. This is a simplification given the variety of smaller early and middle-latency responses that have been found with previous depth electrode recordings from

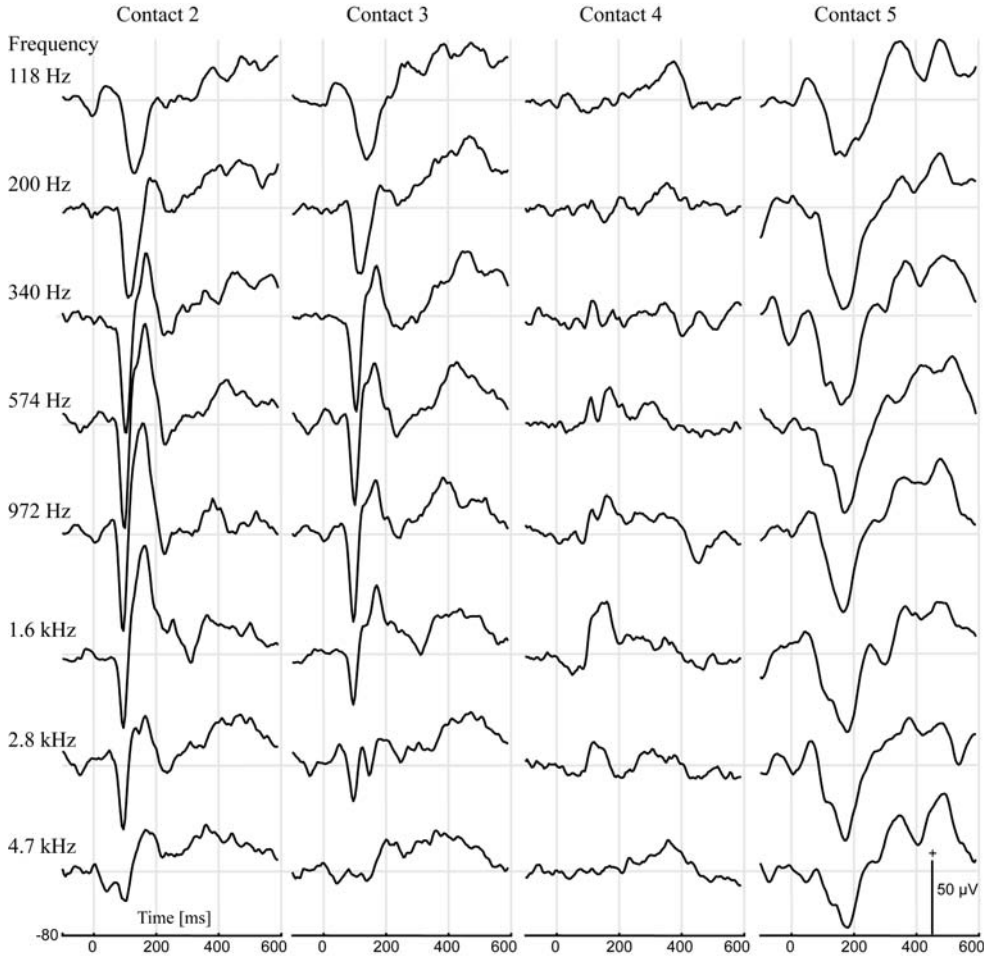

Fig. 3 Responses to pure-tones of eight different frequencies at the four electrode contacts. At contacts *2* and *3*, responses are strongly frequency dependent, while the lateral contact (*5*) shows very similar responses to all frequencies

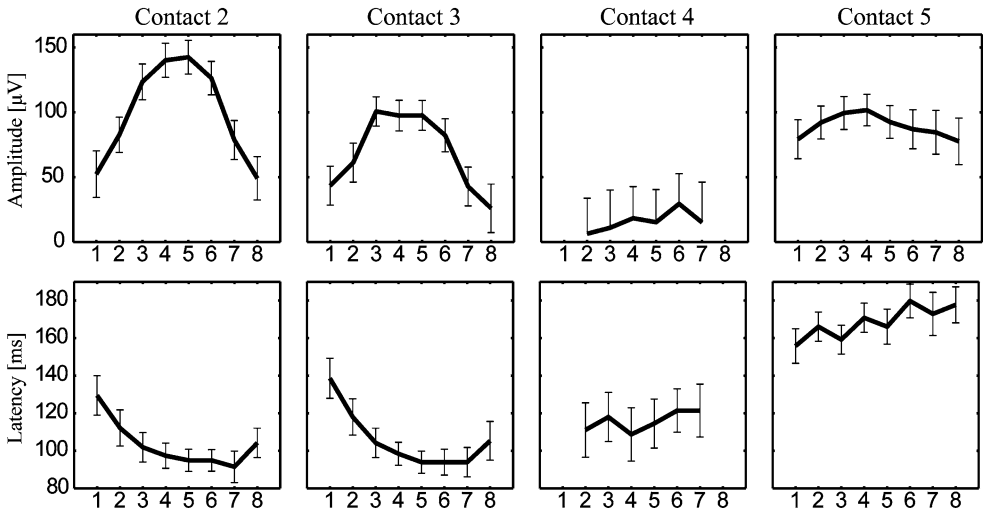

Fig. 4 Peak-to-peak amplitudes (*upper row*) within a time window from 50 to 200 ms past stimulus onset (50–400 ms for contact *5*) and latencies (*lower row*) of the most prominent negative deflection (positive deflection for contact *4*) of the responses to pure tones of eight different frequencies (ordinate, numbered from low to high, frequencies as in Fig. 3) at the four contact sites (*columns*). The two medial contacts show a strong frequency-dependence of both amplitudes and latencies, whereas the amplitude and latency of the responses from the lateral contact *5* is fairly independent of frequency

primary and secondary auditory cortex in man (Celesia 1976; Liegeois-Chauvel et al. 1994, 1991). The present results can nevertheless be compared to the previously reported long-latency pitch-related responses. Second, the depth electrode was placed into possibly epileptogenic tissue the functional properties of which may differ from those of healthy tissue. However, the morphology and latency of auditory responses to clicks were comparable to typical N100 scalp potentials recorded from healthy participants (Näätänen and Picton 1987), suggesting a normal auditory function at the sites of the electrode contacts in our patient. This is an important control given the patient's longstanding epilepsy and evidence of atrophy.

Responses to pure-tones differed markedly between medial and lateral contacts, consistent with the view that the lateral contacts record from secondary auditory cortex. Responses from contact 4 were relatively small and inconsistent for all stimuli, possibly because this contact picks up conflicting response morphologies from neighboring medial and lateral sites. The responses to iterated ripple noise show a clear double dissociation between medial and lateral contact sites. Medial secondary and primary auditory cortex responded strongly to the onset of sound energy, but did not respond to the onset of a pitch that was not accompanied by sound energy changes. Lateral secondary auditory cortex showed the reverse pattern and responded weakly to the sound onset but strongly to the pitch onset. The location of this response complements results form previous neuroimaging studies demonstrating that the lateral secondary auditory cortex extracts pitch from complex sounds.

Electrode placement

We recorded sound evoked potentials from a depth electrode parallel and slightly posterior from Heschl's gyrus. The location of the electrode was determined from the stereotactic coordinates defined during surgical planning. Although no post-implantation MRI was acquired is this case, the stereotactical procedure is accurate to within 1 mm at the tip of the electrode. We were unable to record significant responses from the contact at the electrode tip, which came to lie in white matter at the medial origin of HG. A contact may become unusable after implantation for several reasons, including contact wire failure and immune-reaction coating of the contact surface. Electrical deflections measured at this contact were more than an order of magnitude smaller than those from the neighbouring contact and could not be distinguished from baseline with statistical confidence.

The electrode's wire contacts have low impedance, and are sensitive to the local field potential within an area of about 10 mm radius. From the absence of a response at one contact we can thereby infer that the portion of STP in this area around the contact does not participate in generating the response. Conversely, the presence of a pitch-related response in contact 5, the most lateral contact still within grey matter, indicates that at least part of this response is generated within about 1 cm^3 of brain tissue including the lateral tip of Heschl's gyrus and adjacent superior temporal gyrus.

Response to clicks

The responses to clicks, stimuli that do not evoke a pitch percept, are smallest at the lateral contact and increase in magnitude by a factor of about 2.5 toward the medial contacts. Clicks have been used in previous studies with depth electrodes in the human auditory cortex, and several early (<~30 ms) cortical potentials have been found to originate from the medial two thirds of Heschl's gyrus, probably primary auditory cortex (Celesia and Puletti 1969; Lee et al. 1984; Liegeois-Chauvel et al. 1991; Yvert et al. 2005). After that time, several areas become simultaneously active, leading to high-amplitude long-latency potentials. For instance, Heschl's gyrus, planum temporale, and superior temporal gyrus contribute to the scalp N1 potential (Yvert et al. 2005). We were not able to resolve the relatively weak early primary potentials, due to the location of the electrode adjacent to, but not in, Heschl's gyrus, and the much lower number of stimulus repetitions for averaging compared to studies focussing on early primary potentials (Lee et al. 1984; Liegeois-Chauvel et al. 1991). Nevertheless, we observed a large negative-going wave in the latency range of 95–115 ms at the medial contact—a component of the scalp N1 potential. Negative deflections of similar amplitude and latency have been previously recorded with depth electrodes from this area (Celesia and Puletti 1969; Godey et al. 2001; Yvert et al. 2005). The insensitivity of this potential to pitch and its latency suggest a close relation to the N110m described by Gutschalk and colleagues (2004). The equivalent current dipole of the N110m was localized to the lateral planum temporale; our data suggest a more medial origin within planum temporale.

Response to pure-tones

Pure-tone stimuli evoke a pitch percept, and the onset of the pitch coincides with the onset of sound energy. These stimuli resulted in strong responses from all electrode contacts in HG. The latency and amplitude of responses from the medial contacts depend strongly on stimulus frequency; and the morphology of the response is very similar to the click-evoked response. The variation of the response latency in the medial contacts replicates earlier findings of

latency decreases on the order of 20 ms from low to high frequencies in the N100 from scalp EEG (Woods et al. 1993) and N100m from MEG recordings (Lütkenhöner and Steinsträter 1998; for review see Roberts et al. 2000; Roberts and Poeppel 1996).

Liegeois-Chauvel and colleagues (2001) recorded responses from depth electrodes in HG to pure tones of 8 different frequencies between 250 Hz and 4 kHz. They reported frequency-dependent changes in peak-to-peak amplitudes of the P45/N100 complex very similar to our results. An indication that this frequency-dependence may be related to tonotopic organization comes from microelectrode depth recordings in a single patient by Howard and colleagues (1996), who found differences in spectral tuning of multi-units along the medio-lateral extension of HG.

Responses from the lateral contact do not depend on stimulation frequency and show a very different morphology when compared to the medial responses or the lateral response to clicks. Godey and colleagues (2001) compared responses to clicks and pure-tones measured with MEG and intracranial electrodes in HG and PT. For both types of stimuli they note a change in the morphology of the response (shift to longer latency and broadening of the response peaks) from medial to lateral contacts that is comparable to our observations. The independence of the response latency of stimulus frequency is somewhat surprising because the pitch-onset response to IRN is known to depend systematically on pitch value (Krumbholz et al. 2003). The apparent independence of pure-tone pitch value might have to do with possible differences in the extraction of spectral versus temporal pitch. There is some evidence that lateral Heschl's gyrus responds differently to different types of pitch (but see Chait et al. 2006; Hall and Plack 2007).

The stark changes in the morphology of the responses might be taken as an indication that at least part of the response from the medial contacts comes from tonotopic primary auditory cortex, whereas the lateral contact records from non-tonotopic secondary auditory cortex. The very small responses to clicks and white noise at the lateral contact suggest that the response to pure-tones is mainly due to the tonal character of the stimuli, with only a minor contribution of response to the energy onset.

Pitch onset response

A clear response to the onset of pitch within a noise was found at the electrode contact close to the lateral tip of Heschl's gyrus, but not at the two contacts recording from medial Heschl's gyrus and medial planum temporale. Conversely, the response to the onset of sound energy was larger by a factor of 2–3 at the medial sites than at the lateral site. The pitch onset response reflects synchronized activity from neurons in the auditory cortex that are involved in pitch processing (Krumbholz et al. 2003). The latency of the response depends on the period of the IRN sound and is consistently larger than the latency of the response to a sound energy onset (Krumbholz et al. 2003). In our data, the latency of sound onset responses from the medial contacts varied between 100 and 130 ms for different stimuli, whereas the pitch onset response from the lateral contact had a latency of 220 ms, although the peak is relatively broad. The particular limitations of working with the patient allowed us only to record a relatively low number of repetitions, and consequently there is a considerable amount of noise in the responses. While this does not preclude the detection of a response, it limits the accuracy of amplitude and latency measurements. Although the largest peak of the POR measured here occurred at 220 ms, the response first deviated significantly from the response at contact 2 at a smaller peak with a latency of about 115 ms. The true POR latency probably falls between the two values. Previous MEG experiments, using a higher number of repetitions, have demonstrated latencies of around 150 ms for similar pitch values (Gutschalk et al. 2004; Krumbholz et al. 2003). Those studies also used equivalent current dipole modelling to localize the generator of the pitch onset response approximately to the lateral tip of Heschl's gyrus (Gutschalk et al. 2004; Krumbholz et al. 2003). Similarly, fMRI studies demonstrated pitch-related responses on the lateral Heschl's gyrus and adjacent superior temporal gyrus. Our data confirm this location with depth-electrode recordings from this area, a more direct method than used in previous studies.

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